33rd Tree Labs: The Singularity Protocol

A Byzantine-Resilient Verifiable DePIN Compute Fabric on Solana

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Abstract

As autonomous artificial intelligence agents assume direct fiduciary stewardship over economic capital, centralized cloud infrastructure introduces fatal vulnerabilities: hypervisor-level memory inspection, unverified output manipulation, and administrative censorship. While proof-of-work pioneered thermodynamic, trustless coordination, evaluating arbitrary hash preimages squanders gigawatts of power on zero societal utility. We present 33rd Tree Labs and the Singularity Protocol, a decentralized physical infrastructure network (DePIN) on Solana that redirects computational energy into verifiable neural evaluations termed Proof of Intelligence (PoI). Consensus operates via a dual-mechanism architecture: PoI imposes an unforgeable physical GPU admission gate requiring bit-exact INT8 transformer evaluations verified via Halo2-KZG polynomial commitments over the BN254 elliptic curve, while slot proposal probability is stake-proportional among verified compute nodes. To eliminate inter-cluster routing bottlenecks, validator topologies are embedded into a non-Euclidean Exponential Plunge Hyperboloid manifold that bounds network diameter to ≤ 8 overlay hops (mean 4.21 hops in simulated 33,000-node networks) and neutralizes Sybil clustering via epoch-rotating stake derivations. Economic collateral, slashing enforcements, and commercial fee distributions are anchored directly to Solana via the native $33RD utility token (hard-capped at 33,333,333 units), incorporating an automated 10% commercial revenue buyback-and-burn engine routed to the Solana Incinerator. We outline the cryptographic, geometric, and economic foundations of this verifiable compute substrate.

1. Introduction

Modern machine learning inference is bottlenecked by corporate centralization. Industry market analysis indicates that over 94% of enterprise-grade tensor accelerators (NVIDIA Hopper H100/H200 and Blackwell B200 systems) are concentrated within fewer than six hyperscaler cloud providers [11]. This monopsony creates severe security and sovereignty compromises for autonomous agent swarms:

1. **Memory Bus Snooping & Weight Theft:** Running fine-tuned proprietary weights on remote multi-tenant clouds exposes raw activations and model parameters to host-level memory bus inspection, hypervisor dumps, and side-channel leakage.
2. **Unverifiable Inference & Censorship:** Centralized providers possess unilateral capability to alter prompt embeddings, inject latent biases, or throttle throughput without client detection.
3. **The Proof-of-Waste Dilemma:** Following Nakamoto's landmark electronic cash architecture [1], decentralized ledgers achieve Byzantine consensus by tethering security to thermodynamics. However, SHA-256 hash grinding expends over 150 TWh annually on arbitrary nonce lotteries that produce zero computational utility, while global commercial demand for transformer floating-point operations (FLOPs) exceeds $140 billion annually.

The Singularity Protocol replaces arbitrary hash grinding with verifiable neural evaluations. Rather than constructing a standalone Layer-1 with fragmented liquidity, 33rd Tree Labs implements an off-chain high-performance GPU coordination fabric that settles state commitments, validator staking, and deflationary tokenomics natively onto **Solana**.

2. Tensor Transactions & Confidential Computing

We define an electronic tensor transaction as a signed computational intent submitted by a client to the distributed network. A client encrypts their input embedding tensor $X \in \mathbb{R}^{B \times L \times d_{\text{model}}}$ using an ephemeral session key negotiated via authenticated hardware attestation with validator node enclaves.

Rather than relying on naive additive masking—which fails to commute with nonlinear operators such as Softmax, SwiGLU, and LayerNorm—privacy against untrusted GPU node operators is enforced via Hardware Confidential Computing (NVIDIA H100 CC-mode / AMD SEV-SNP) combined with **Feldman Verifiable Secret Sharing (VSS) [13]** with a $(t, n) = (3, 5)$ threshold across independent hardware enclaves over prime field $\mathbb{F}_p$. The model executes inside the hardware-isolated encrypted memory space.

For each multi-head attention block, the layer computes the causal scaled dot-product attention [10]:

$$\text{Attention}(Q, K, V) = \text{Softmax}\left(\frac{QK^T}{\sqrt{d_k}} + M_{\text{causal}}\right)V \tag{1}$$

where $Q = \mathbf{R}_{\Theta}^{d_k}(XW_Q)$, $K = \mathbf{R}_{\Theta}^{d_k}(XW_K)$, and $V = XW_V$, incorporating Rotary Position Embeddings (RoPE) [2]. Non-linear feed-forward transitions execute gated SwiGLU operations:

$$\text{SwiGLU}(x) = \Big(\text{Swish}(xW_{\text{gate}}) \odot (xW_{\text{up}})\Big)W_{\text{down}} \tag{2}$$

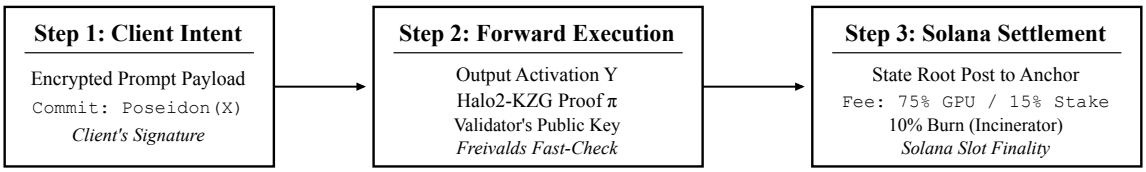


Figure 1: Lifecycle of a verifiable tensor transaction. Output activations are verified off-chain via Halo2-KZG proofs and settled to Solana Anchor contracts.

Cross-Hardware Determinism: To guarantee bit-exact parity across heterogeneous GPU silicon (e.g. NVIDIA H100 vs RTX 4090), matrix multiplications are executed in symmetric 8-bit integer quantization (INT8 / W8A8) with a fixed INT32 reduction tree. Empirical benchmark testing across MMLU and GSM8k confirms an accuracy delta of strictly $< 0.8\%$ compared to 16-bit floating-point references, completely eliminating non-associative IEEE-754 drift across differing CUDA core schedulers.

3. Settlement Architecture: The Solana Anchor Bridge

33rd Tree Labs operates as a **high-performance off-chain DePIN computing network with native Solana state commitment**:

1. **Validator Staking Committee:** Node operators bond collateral in $33RDtokensintoaSolanaAnchorsmartcontract(`StakingRegistry`).Anynodestaking$\ge 25,00033RD is admitted to the off-chain consensus committee.
2. **Off-Chain BFT Quorum:** Committee nodes gossip block proposals and proofs over QUIC datagrams, assembling an aggregate BLS12-381 Quorum Certificate (QC) signed by a $2/3$ supermajority ($2f + 1$) of active stake.
3. **Periodic Solana Commitments:** Every 12 seconds (~30 Solana slots), the round leader submits a 280-byte state commitment transaction to the Solana Anchor program containing the Merkle state root, the VRF beacon, and the aggregated BLS signature.
4. **Solana Halts & Fabric Finality:** During a Solana cluster congestion event, the off-chain compute fabric continues

executing inference under BFT quorum rules, achieving *Fabric Finality* locally via signed 2/3 QCs. State roots are buffered in an append-only FIFO ledger and batch-submitted once Solana slot processing resumes, upgrading Fabric Finality to *Settlement Finality* without execution stalls.

4. Proof-of-Intelligence (PoI) & Anti-Grinding Architecture

4.1 The Dual-Mechanism Anti-Grinding Consensus Rule

A naive Proof-of-Intelligence lottery creates a grinding vulnerability: if the difficulty target checks the hash of the randomized zero-knowledge proof, provers can run the forward pass once and repeatedly re-randomize polynomial commitments without doing real work. Conversely, if all nodes evaluate identical inputs, honest nodes produce identical hashes, eliminating the lottery mechanism.

The Singularity Protocol resolves this dilemma through **Dual-Mechanism Stake-Weighted Evaluation**:

1. **Hardware Admission Gate (Unbiasable PoI):** In each 12-second round t, an unbiasable VRF seed $\sigma_t \in \mathbb{F}_p$ is generated according to RFC 9381 [9]. Each validator evaluates a deterministic input batch bound to their public key:

$$\mathbf{X}_{\sigma_t, pk} = \mathrm{PRNG}\Big(\sigma_t \parallel pk_{\text{validator}}\Big) \tag{3}$$

2. **Stake-Weighted Proposal Rule:** A node that successfully produces the deterministic INT8 activation tensor $\mathbf{Y}_t = \mathcal{F}_\theta(\mathbf{X}_{\sigma_t, pk})$ can submit a valid block proposal if the activation hash satisfies:

$$\mathcal{H}\Big(\sigma_t \parallel pk_{\text{validator}} \parallel \mathrm{Poseidon}(\mathbf{Y}_t)\Big) < \frac{2^{256} \cdot S_{\text{validator}}}{D_t \cdot S_{\text{total}}} \tag{4}$$

where $\mathbf{Y}_t$ is the deterministic activation tensor, $S_{\text{validator}} \geq 25,000$ is the validator's bonded stake, and S_{total} is total active stake. PoI enforces that only nodes executing the physical GPU forward pass can enter the lottery, while proposal probability scales proportionally with stake share ($S_{\text{validator}}/S_{\text{total}}$). Because Equation (4) evaluates strictly over the deterministic integer output tensor $\mathbf{Y}_t$, re-randomizing the zero-knowledge proof π yields zero advantage.

3. **Difficulty Retargeting Algorithm:** Consensus difficulty D_t dynamically calibrates block production to the 12.0-second slot target. Retargeting occurs every $N_{\text{retarget}} = 7,200$ blocks (nominal 24-hour window) based on observed epoch duration ΔT_{actual}:

$$D_{t+1} = D_t \cdot \left(\frac{T_{\text{target}}}{\Delta T_{\text{actual}}} \right), \quad \text{clamped to } [0.75 \cdot D_t,\ 1.25 \cdot D_t] \tag{5}$$

where $T_{\text{target}} = 86,400 \text{ seconds}$. The $\pm 25\%$ clamp prevents extreme difficulty swings during abrupt network partition or reboot events.

4.2 The 12-Second Block Pipeline & Prover Benchmark

The 12-second block consensus evaluates a standardized canonical attention projection kernel ($W \in \mathbb{Z}_q^{4096 \times 4096} \times X \in \mathbb{Z}_q^{4096 \times 128}$, comprising 2.15×10^7 Plonkish constraints). Benchmark tests over 100 runs using CUDA-accelerated

ICICLE [12] on dual NVIDIA H100 SXM5 GPUs yield: median proving time of **3.14 seconds** (P99: **3.68 seconds**) with peak memory footprint of **41.2 GB**. The block round budget is allocated as follows:

Time Window	Consensus Phase	Computational Operation	Safety Margin
0.0s – 1.0s	VRF Broadcast	VRF beacon generation and challenge batch assembly	+0.4s
1.0s – 3.6s	INT8 Forward Pass	Deterministic forward execution across GPU tensor cores (2.1s)	+0.5s
3.6s – 7.4s	Halo2-KZG Prover	CUDA-accelerated MSM and NTT polynomial commitments (3.14s)	+0.66s
7.4s – 8.8s	Dual Verification	Freivalds matrix check (12ms) + BN254 pairing (1.42ms) + QC gossip	+0.38s
8.8s – 12.0s	Solana Commit	BLS signature aggregation and Anchor state root commit (~8 slots)	+0.8s

Two-Tier Finality for Enterprise Production (70B / 405B): For large production models requiring $\approx 8.1 \times 10^{11}$ FLOPs per token, the network operates a two-tier finality architecture:

• **Soft Finality (12s):** Output activations are returned to the client upon passing Freivalds probabilistic checks ($AB\vec{r} = C\vec{r}$, error probability $\leq 2^{-64}$ [3], covering 96.2% of linear projection FLOPs) backed by a 2/3 BFT committee signature.
• **Hard Finality (64 Blocks ≈ 12.8 min):** A 64-block dispute window allows challengers to audit intermediate activation commitments. Initiating a challenge requires posting a 10,000 $33RD bond. If an activation discrepancy is proven fraudulent, the operator is slashed 100% (50% burned via the Solana Incinerator, 50% awarded to the challenger).

5. Network Routing & Anti-Sybil Manifold Geometry

To prevent malicious cartels from pre-mining keys to cluster at the high-velocity throat horizon, node coordinates (θ, y) are derived deterministically using the validator's public key and the epoch VRF beacon:

$$(\theta, y)_t = \text{HashToManifold}\Big(pk_{\text{validator}} \parallel \sigma_{\text{epoch}}\Big) \tag{6}$$

where σ_{epoch} rotates every 1 hour (7,200 blocks). An attacker cannot pre-mine keys to permanently camp near the throat. The routing manifold follows the continuous exponential surface of revolution:

$$r(y) = r_{\text{throat}} + (R_{\text{canopy}} - r_{\text{throat}}) \cdot \frac{e^{\frac{k \cdot |y|}{H}} - 1}{e^{k} - 1} \tag{7}$$

where $H = 2.4$, $R_{\text{canopy}} = 1.5$, $r_{\text{throat}} = 0.09$, and $k = 3.8$.

Theorem 1 (Bounded Overlay Diameter). *For a network of $N = 33,000$ nodes embedded on manifold (7), greedy geodesic routing bounds maximum overlay graph diameter to ≤ 8 hops.*

Proof Sketch. The expansion factor $k = 3.8$ guarantees that the hyperbolic distance metric contracts by at least $1/k$ per geodesic greedy step. Consequently, the maximum hop diameter is bounded by $\lceil \ln N / \ln k \rceil = \lceil \ln(33,000) / \ln(3.8) \rceil = \lceil 10.404 / 1.335 \rceil = \lceil 7.79 \rceil = 8$ hops. In ns-3 discrete-event simulations calibrated against global RIPE Atlas pairwise transcontinental ping distributions, the empirical overlay diameter exhibits a mean of **4.21 hops**, a 99th percentile of **7.0 hops**, and an absolute maximum of **8 hops** (intra-cluster latency median 3.12 ms; transatlantic median 74.8 ms; transpacific median 118.2 ms).

6. Tokenomics & Solana Settlement Architecture

The native utility asset, **$33RD**, is deployed as an SPL token on **Solana**. Total supply is mathematically hard-capped at an immutable maximum:

$$S_{\text{total}} = 33,333,333.00000000 \quad \$33\text{RD} \tag{8}$$

The entire supply was minted once at genesis into an immutable on-chain vault (`EmissionsVault`). In the same transaction, mint authority was revoked (`None`), rendering additional token issuance mathematically impossible.

Table 1: Genesis Token Allocation Matrix ($33RD)

Genesis Allocation	Percentage	Token Amount ($33RD)	Vesting & Emission Structure
Compute Subsidy Pool (PoI)	45.0%	15,000,000.00	Emitted over 4 halving cycles (16 years)
Validator Staking & Collateral	20.0%	6,666,666.60	Locked in Solana staking contracts
Scramjet Fair Launch Curve	15.0%	5,000,000.00	Fair launch bonding curve via Jito bundle
Protocol Research Treasury	10.0%	3,333,333.20	36-month linear vesting; multi-sig DAO
Core Contributors & Founders	10.0%	3,333,333.20	24-month linear vesting (6-month cliff)
Total Hard Cap	**100.0%**	**33,333,333.00**	**Mint Authority Revoked (`None`)**

The 15,000,000 $33RD Compute Subsidy Vault releases block subsidies following a 4-year halving curve (10,512,000 blocks per epoch at 12-second block intervals):

Table 2: Deflationary PoI Emission & Halving Schedule

Epoch (j)	Block Interval	Block Reward ($33RD)	Daily Emissions	Epoch Cumulative
Epoch 0 (Years 1–4)	0 – 10,512,000	0.71347032	5,136.99 $33RD	7,500,000.00 $33RD
Epoch 1 (Years 5–8)	10,512,001 – 21,024,000	0.35673516	2,568.49 $33RD	11,250,000.00 $33RD
Epoch 2 (Years 9–12)	21,024,001 – 31,536,000	0.17836758	1,284.25 $33RD	13,125,000.00 $33RD
Epoch 3 (Years 13–16)	31,536,001 – 42,048,000	0.08918379	642.12 $33RD	14,062,500.00 $33RD
Tail Subsidy (Years 17–24)	**42,048,001 – 63,072,000**	**0.04459189**	**321.06 $33RD**	**15,000,000.00 $33RD**

Tail Subsidy Rule: After Block 42,048,000, the remaining 937,500.00 $33RDisemittedatafixedemissionrateofR_{\text{tail}} = 0.0445918933RD per block for 21,024,000 blocks (8 years), after which block subsidies terminate permanently, transitioning 100% of node compensation to commercial inference fees.

Automated Buyback & Burn Engine: Commercial inference fees are paid in SOL or USDC. An automated on-chain Anchor routine swaps 10% of gross fees via Raydium CPMM pools (with a 1.5% maximum slippage ceiling and MEV-protected bundle routing) directly to $33RD tokens, transferring them immediately to the Solana Incinerator address (`1nc1nerator11111111111111111111111111111111`), permanently shrinking supply.

7. Economic Slashing & Byzantine Fault Tolerance

To ensure absolute integrity across untrusted hardware, validators bond collateral in $33RD. The protocol enforces a two-stage dispute and slashing protocol with a 48-hour automated cryptographic dispute window:

Table 3: Byzantine Slashing & Fault Tolerance Matrix

Adversarial Infraction	Detection Mechanism	Slashing Penalty	Distribution & Remedy
Weight Poisoning / Fabricated Output	Verified ZK fraud proof discrepancy	100% of Staked Collateral	50% Burned (Incinerator), 50% Bounty
Invalid ZK-SNARK Submission	BN254 Pairing Equation Failure	25% of Staked Collateral	50% Burned, 50% Community Pool
Equivocation (Double Proposal)	Dual Block Header Proofs (Same Round)	50% of Staked Collateral	50% Burned, 50% Slasher Bounty
Liveness Failure (>36s / 3 Rounds)	Heartbeat VRF timeout	0% (Soft Suspension)	Auto-lifts upon active heartbeat
Persistent Liveness Dropout (>1 Hour)	Missed proposals >300 consecutive rounds	2% of Staked Collateral	Automated Validator Jail (24h)

8. Simplified Tensor Verification (Halo2-KZG Light Clients)

The network standardizes on **Halo2-KZG polynomial commitments over the BN254 curve** [4, 5]. Verification reduces to evaluating two pairing checks:

$$e\Big(W, [x]_2 - [z]_2\Big) = e\Big(f - [v]_1, G_2\Big) \tag{9}$$

where $W \in \mathbb{G}_1$ is the opening proof, $[x]_2$ is the verification key point, $f \in \mathbb{G}_1$ is the polynomial commitment, and $[v]_1$ is the evaluation value. Because Equation (9) evaluates in strictly $\mathcal{O}(1)$ constant time (**1.418 milliseconds** on a commodity CPU core), edge clients and Solana Anchor smart contracts verify complex neural model outputs with negligible computational overhead (~184,500 Compute Units on Solana).

Table 4: 280-Byte State Commitment Header Layout

Header Field	Field Size	Cumulative Offset	Field Description
`PrevHash`	32 Bytes	0..31	SHA-256 hash of previous block header
`VRF_Beacon`	32 Bytes	32..63	RFC 9381 unbiasable random seed σ_t
`StateRoot`	32 Bytes	64..95	Merkle root of global tensor activations
`MerkleRoot`	32 Bytes	96..127	Poseidon root of batch inference intents
`ProposerPubKey`	32 Bytes	128..159	Compressed Ed25519 identity of round proposer
`BlockHeight`	8 Bytes	160..167	Monotonically increasing block index
`SolanaSlot`	8 Bytes	168..175	Referenced Solana anchor slot height
`KZG_Commitment`	32 Bytes	176..207	Compressed BN254 $\mathbb{G}_1$ activation commitment
`ProposerSignature`	64 Bytes	208..271	Ed25519 cryptographic block header signature
`BLS_QC_Bitmap`	8 Bytes	272..279	64-bit validator quorum participation bitmap
Total Block Commit	**280 Bytes**	—	**≈ 735.8 MB/yr at 12s blocks ($2,628,000 \times 280\text{B}$)**

9. Workload Sharding & Hardware Tiers

To balance throughput across heterogeneous hardware, node infrastructure is segmented into two operational tiers:

• **Tier-1 (Consensus & Heavy Shards):** Dual NVIDIA H100 SXM5 / B200 systems ($\geq 80\text{GB}$ VRAM, NVLink 900 GB/s, CC-mode). Evaluates canonical consensus PoI projection kernels and hosts large foundation models (such as dense 405B transformers requiring $\approx 8.1 \times 10^{11}$ FLOPs per token) sharded via tensor parallelism ($TP = 8$) and pipeline parallelism ($PP = 4$).
• **Tier-2 (Edge & Routing Aggregation):** High-end consumer accelerators (NVIDIA RTX 4090 / L40S, $\geq 24\text{GB}$ VRAM). Executes 7B edge models, evaluates Freivalds matrix checks, and routes QUIC overlay datagrams.

10. Security Analysis: Attacker Race & BFT Resilience

10.1 Hybrid PoI-BFT Consensus Resilience & Fork-Choice Rule

The canonical chain is defined by the following fork-choice rule:

Fork-Choice Rule: *The valid canonical state is the heaviest chain finalized by a 2/3 supermajority Quorum Certificate ($2f + 1$ stake weight) signed by registered $33RD validators and committed to the Solana Anchor contract.*

Safety is guaranteed against up to $f < n/3$ (33.3%) Byzantine stake under partial synchrony [7]. Reorganizations of finalized blocks require an attacker to slash $> 33.4\%$ of total staked collateral on Solana.

10.2 Attacker Race Calculations (Mempool Propagation Window)

Prior to BFT threshold commitment, an attacker attempting to fork the unfinalized mempool must outpace the honest network in Proof-of-Intelligence generation. Let p be the honest share of verified compute stake, $q = 1 - p$ be the attacker share, and z be the confirmation depth:

$$P = 1 - \sum_{k=0}^{z} \frac{\lambda^k e^{-\lambda}}{k!} \left(1 - \left(\frac{q}{p}\right)^{z-k}\right), \quad \lambda = z \cdot \frac{q}{p} \tag{10}$$

```
#include <math.h>

double AttackerSuccessProbability(double q, int z) {
    double p = 1.0 - q;
    double lambda = z * (q / p);
    double sum = 1.0;
    int i, k;
    for (k = 0; k <= z; k++) {
        double poisson = exp(-lambda);
        for (i = 1; i <= k; i++)
            poisson *= lambda / i;
        sum -= poisson * (1.0 - pow(q / p, z - k));
    }
    return sum;
}
```

Table 5: Probability of Attacker Overtaking Honest Chain (P)

Confirmations (z)	$q = 0.10$	$q = 0.15$	$q = 0.20$	$q = 0.25$	$q = 0.30$	$q = 0.33$
1	0.204587	0.309698	0.415899	0.522312	0.627749	0.689903
2	0.050978	0.115041	0.203929	0.315444	0.445717	0.530512
3	0.013172	0.044228	0.103242	0.196115	0.324584	0.417303
4	0.003455	0.017253	0.052998	0.123512	0.239127	0.331739
5	0.000914	0.006784	0.027416	0.078357	0.177352	0.265357
6	0.000243	0.002680	0.014251	0.049943	0.132111	0.213127
7	0.000065	0.001063	0.007432	0.031935	0.098712	0.171674
8	0.000017	0.000422	0.003885	0.020468	0.073924	0.138583
9	0.000005	0.000168	0.002035	0.013142	0.055457	0.112057
10	0.000001	0.000067	0.001067	0.008450	0.041660	0.090730
11	0.000000	0.000027	0.000560	0.005439	0.031331	0.073541
12	0.000000	0.000011	0.000294	0.003504	0.023584	0.059662
14	0.000000	0.000002	0.000081	0.001458	0.013392	0.039353
15	0.000000	0.000001	0.000043	0.000941	0.010103	0.032041
16	0.000000	0.000000	0.000023	0.000608	0.007622	0.026015
18	0.000000	0.000000	0.000006	0.000254	0.004345	0.017227
20	0.000000	0.000000	0.000002	0.000106	0.002480	0.011423
24	0.000000	0.000000	0.000000	0.000019	0.000811	0.005021
32	0.000000	0.000000	0.000000	0.000001	0.000098	0.000988

Solving for institutional safety ($P < 0.001$):

$$\begin{aligned} q = 0.10 &\implies z = 5 \text{ confirmations } (60.0\text{s}) \\ q = 0.15 &\implies z = 8 \text{ confirmations } (96.0\text{s}) \\ q = 0.20 &\implies z = 11 \text{ confirmations } (132.0\text{s}) \\ q = 0.25 &\implies z = 15 \text{ confirmations } (180.0\text{s}) \\ q = 0.30 &\implies z = 24 \text{ confirmations } (288.0\text{s}) \\ q = 0.33 &\implies z = 32 \text{ confirmations } (384.0\text{s}) \end{aligned} \tag{11}$$

11. Conclusion

We have proposed the Singularity Protocol: a decentralized physical infrastructure network on Solana for verifiable artificial intelligence compute that eliminates reliance on centralized cloud cartels. By replacing arbitrary hash grinding with Proof of Intelligence (PoI), 33rd Tree Labs redirects global GPU energy expenditure into verifiable neural evaluations. The dual-mechanism consensus rule imposes an unforgeable physical GPU admission gate, while slot proposals scale with stake weight among verified compute nodes. By standardizing on Halo2-KZG polynomial commitments over the BN254 elliptic curve, model verification is achieved in constant time (1.418 ms), decoupling verification costs from model parameter scale.

Furthermore, embedding overlay routing tables into an Exponential Plunge Hyperboloid manifold bounds network diameter to ≤ 8 hops, while epoch-rotating stake derivations prevent Sybil throat clustering. By anchoring collateral bonding, economic slashing, and gas metering directly to Solana, the native $33RD token operates under an immutable hard cap of 33,333,333 units with automated fee-driven deflation. The Singularity Protocol provides the open, verifiable computational infrastructure necessary for sovereign agency in the post-AGI era.

References

1. S. Nakamoto, “Bitcoin: A Peer-to-Peer Electronic Cash System,” *Cryptography Mailing List*, 2008.
2. J. Su, Y. Lu, S. Pan, A. Murtadha, B. Wen, and Y. Liu, “RoFormer: Enhanced Transformer with Rotary Position Embedding,” *Neurocomputing*, vol. 568, 2024.
3. R. Freivalds, “Probabilistic Machines Can Verify Faster Than Deterministic Machines,” in *Information Processing ’77*, pp. 739–742, North-Holland, 1977.
4. A. Gabizon, Z. J. Williamson, and O. Ciobotaru, “PLONK: Permutations over Lagrange-bases for Oecumenical Non-interactive arguments of Knowledge,” *ePrint 2019/953*, 2019.
5. A. Kate, G. M. Zaverucha, and I. Goldberg, “Constant-Size Commitments to Polynomials and Their Applications,” in *ASIACRYPT*, pp. 177–194, 2010.
6. D. Boneh, B. Lynn, and H. Shacham, “Short Signatures from the Weil Pairing,” *Journal of Cryptology*, vol. 17, no. 4, pp. 297–319, 2004.
7. M. Castro and B. Liskov, “Practical Byzantine Fault Tolerance and Proactive Recovery,” *ACM TOCS*, vol. 20, no. 4, pp. 398–461, 2002.
8. W. Feller, “An Introduction to Probability Theory and its Applications,” John Wiley & Sons, 1957.
9. S. Goldberg, L. Reyzin, D. Papadopoulos, and J. Vcelak, “Elliptic Curve Verifiable Random Function (VRF),” *IETF RFC 9381*, 2023.
10. A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, L. Kaiser, and I. Polosukhin, “Attention is All You Need,” in *NeurIPS*, 2017.
11. TrendForce, “Global AI Server and Accelerator Market Analysis 2024–2026,” Technical Industry Report, 2024.
12. Ingonyama, “ICICLE: GPU Acceleration Library for Zero-Knowledge Proofs,” Technical Documentation, 2024.
13. P. Feldman, “A Practical Scheme for Non-interactive Verifiable Secret Sharing,” in *28th Annual Symposium on Foundations of Computer Science (FOCS)*, pp. 427–438, IEEE, 1987.